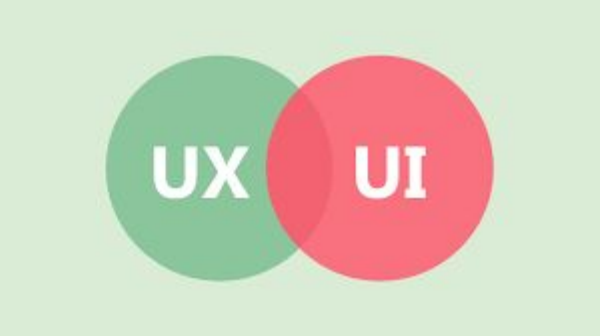
Integrated MSc(IT)

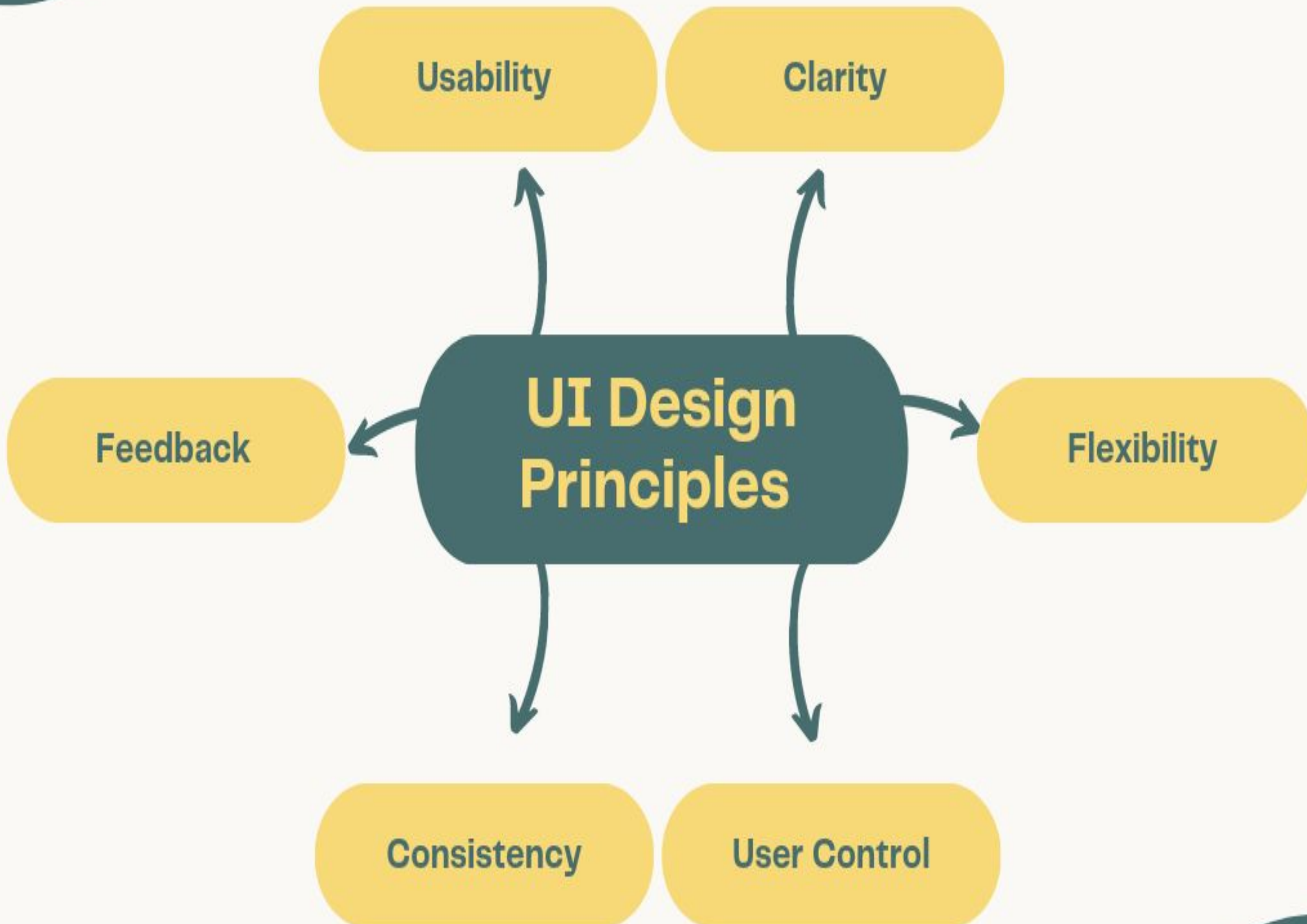
UNIT – III
UI/ UX Design &
Coding, Documentation Standards

221601603 Software Engineering

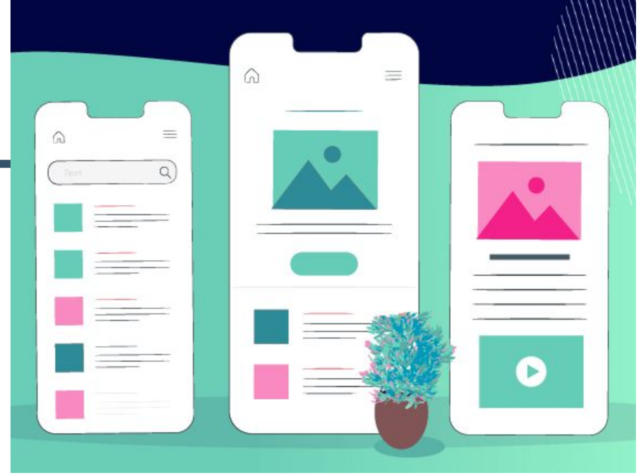
UI/ UX Design



- UI stands for user interface, and UX stands for user experience.
- UI and UX are two related but distinct concepts that are often used in the tech industry:
- **UI:**
  - The visual elements of a product, such as buttons, icons, and page layouts.
  - UI designers focus on the look and feel of a product, and how users interact with it.
- **UX**
  - The overall experience a user has with a product, including how they feel about it.
  - UX designers consider the user's needs, goals, and pain points to create products that are easy, efficient, and enjoyable to use.
- While UI and UX are different, they are both important for creating a positive user experience.
- UI and UX designers often work together, with UX designers creating wireframes and prototypes, and UI designers creating mockups and style guides.

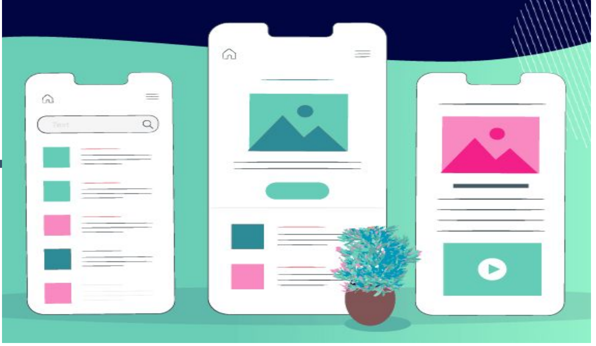

UI Design Principles
Usability
Clarity
Feedback
Flexibility
Consistency
User Control

UI Design Principles



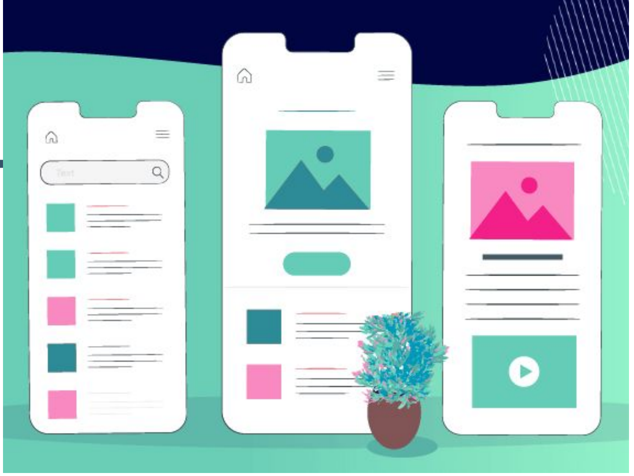
- **Usability**:
  - One of the most important principles of UI design is usability.
  - A user-friendly interface is one that is easy to understand, navigate, and use.
  - This means that the interface should be organized in a logical and intuitive way, with clear and consistent labeling and navigation.
  - Users should be able to accomplish their tasks quickly and easily, without having to learn a lot of new information or navigate through a complex interface.
- **Clarity:**
- A clear interface is one that is easy to read and understand, with a simple and uncluttered layout.
- This means that designers should use a limited color palette, simple typography, and plenty of white space to create a clean and uncluttered look.
- Additionally, designers should ensure that the interface is easy to read by using high-contrast colors, large text, and a consistent font throughout the interface.

UI Design Principles



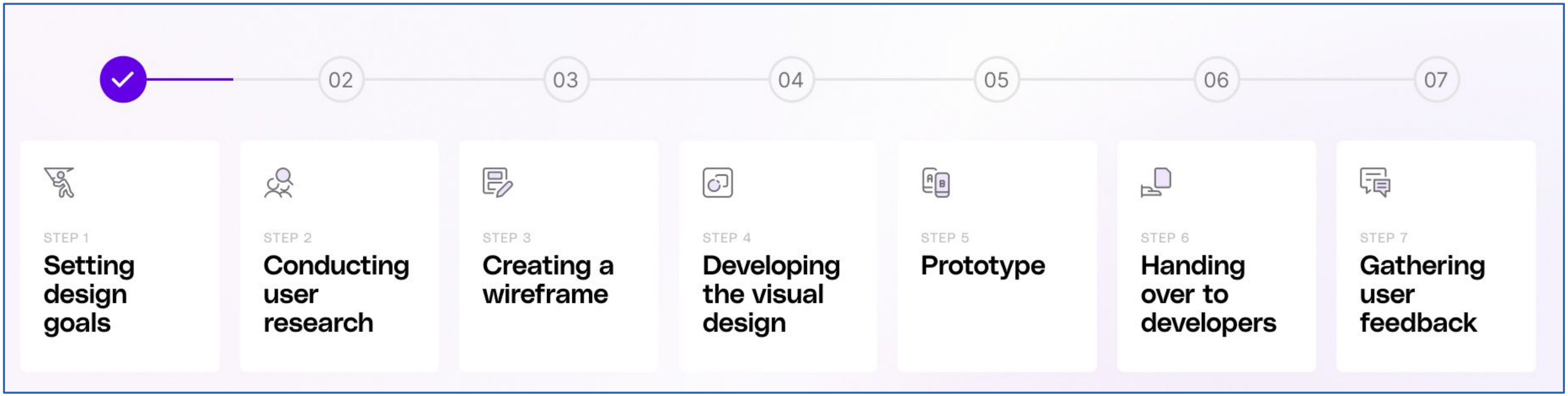
- **Flexibility**
  - Flexibility is an important principle of UI design because it allows users to customize the interface to suit their needs.
  - This means that designers should create interfaces that are responsive, so that they can adapt to different screen sizes and resolutions.
  - Additionally, designers should provide users with the ability to customize the interface, such as by allowing them to change the color scheme or layout.
- **Consistency**
  - Consistency is another important principle of UI design, as it helps users navigate the interface more easily.
  - This means that designers should use consistent layout, typography, and color schemes throughout the interface.
  - Additionally, designers should ensure that the interface is consistent with other interfaces that users are likely to have encountered, such as those in the operating system or web browser.

UI Design Principles



- **Efficiency:**
  - Consistent UI designs promote efficiency.
  - Users don’t have to spend time relearning or searching for information or functionality if the design remains consistent.
  - This saves time and allows users to accomplish tasks more quickly and efficiently.
- **Feedback**
  - Feedback is the principle of UI design that helps users understand what is happening on the interface.
  - This means that designers should provide users with feedback on their actions, such as by highlighting selected buttons or displaying error messages when something goes wrong.
  - Additionally, designers should provide users with visual cues, such as animations and icons, to help them understand the interface more easily.

UI Design Process



UI Design Process: Set Design Goals / Objectives

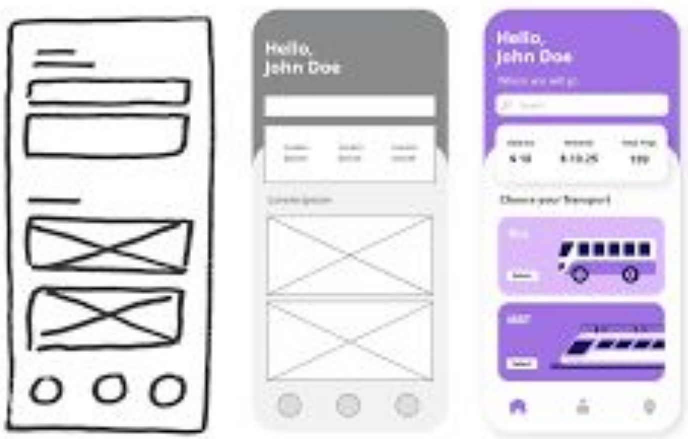
- The first step in the UI design process is to define the goals your team aims to achieve.
- Design solves problems, so understanding the problem is crucial before providing a solution.
- Design teams rely on design briefs—a document outlining all essential components of the design project.
- The **Design Brief** serves as the project's backbone and a communication tool between the customer and designer, offering insights into user needs and the type of design that will fulfill those needs.
- Clearly state the purpose of the design.
- Detail what the project will cover.
- Specify whether it's a new website or application or enhancements to an existing product.

UI Design Process: User Research

- User experience research is the systematic investigation of your users in order to gather insights that will inform the design process.
- It enables to understand users’
  - needs
  - attitudes
  - pain points
  - behaviors
- Typically done at the start of a project—but also extremely valuable throughout—it encompasses different types of research methodology to gather both qualitative and quantitative data in relation to your product or service.

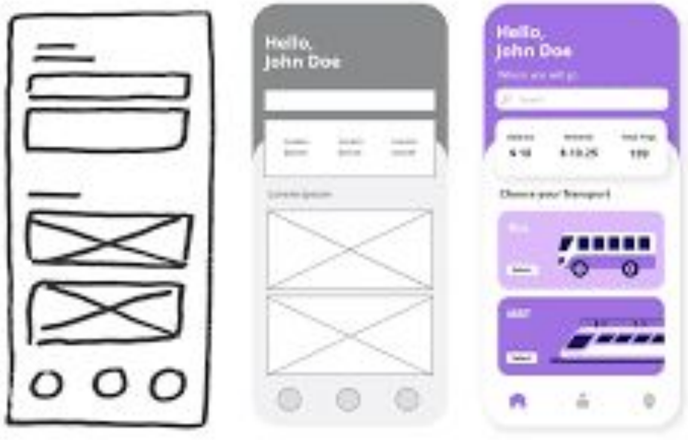
UI Design Process: Wire Framing

- Wireframing is the process of creating a basic layout of a website, app, or software interface before the design and development phase.
- Wireframes are created before actually jumping on the final UI design.
- This step of the process focuses on giving shapes to the rough ideas we collected from concept sketching.
- They outline the structure and functionality of your product without getting caught up in colors, visuals, or specific content.
- While wireframing text, sample relatable images, and dummy content such as lorem ipsum can be added.



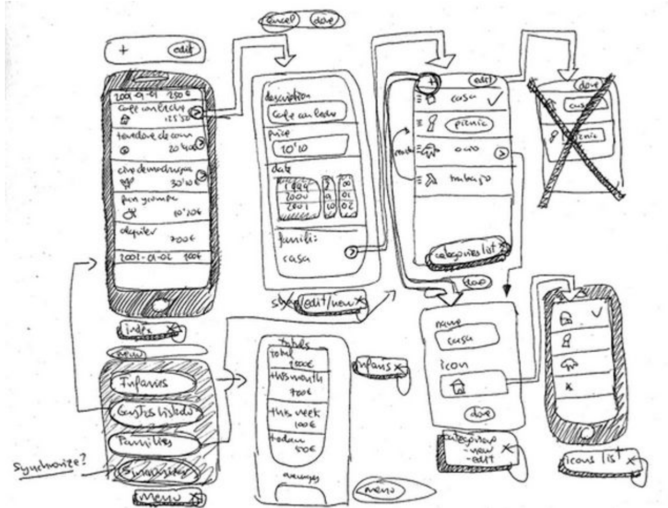
UI Design Process: Wire Framing

- **Benefits of Wireframing:**
  - Helps refine ideas.
  - Helps to visualize the layout
  - and elements of the screen
  - A fast and cost-efficient way of
  - designing and getting feedback
  - Helps in presenting ideas among stakeholders
  - Encourages elimination of irrelevant
  - components and ideas in design



UI Design Process: Wireframing

- A wireframe is a two-dimensional skeletal outline of a webpage or app.
- Wireframes provide a clear overview of the page structure, layout, information architecture, user flow, functionality, and intended behaviors.
- Styling, color, graphics, and other design elements are kept to a minimum.
- They can be drawn by hand or created digitally, depending on how much detail is required.
- Wireframing is a practice most commonly used by UX designers.
- This process allows all stakeholders to agree on where the information will be placed before the developers build the interface out with code.
- Wireframing facilitates feedback from users, instigate conversations with stakeholders, and generate ideas between the designers.

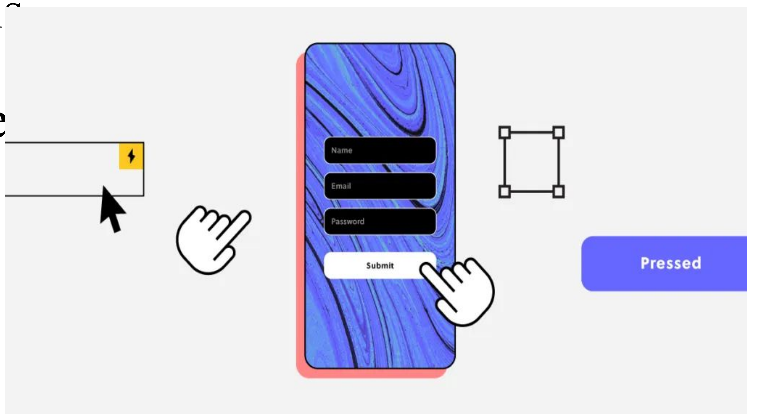


UI Design Process: Developing Visual Design

- Visual design aims to improve a design's/product's aesthetic appeal and usability with suitable images, typography, space, layout and color.
- Visual design is about more than aesthetics.
- Designers place elements carefully to create interfaces that optimize user experience and drive conversion.
- Visual design goes beyond making layouts and designs look nice and aesthetically pleasing.
- It can increase usability, provoke emotion, and strengthen brand perception.
- Visual design in UX design focuses on making digital products like websites and apps look good and easy to use.
- It combines colors, fonts, shapes, and images to create attractive interfaces that enhance the user experience.
- The goal is to communicate information clearly and guide users smoothly through the product, while also ensuring a visually appealing layout.
- In UX design, strong visual design helps engage users, improves usability, and makes navigating the website or app more enjoyable and efficient.

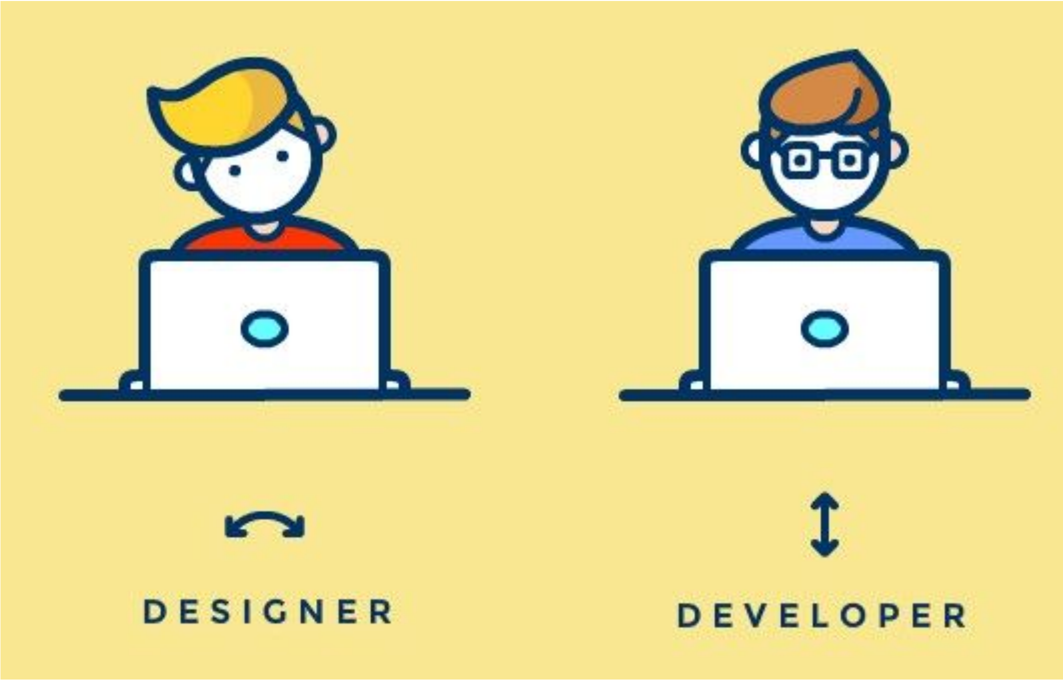
UI Design Process: Prototype

- A prototype is an early model or simulation of a product used to test and validate ideas before full-scale production.
- A prototype is a representation of the end-product that is used in order to see if the product teams are building the right solution for their desired users.
- Prototypes vary in fidelity from simple sketches of a user interface to fully interactive digital models that resemble the final product.
- They serve to gather user feedback, identify usability issues, and refine design concepts, helping ensure that the final product meets user needs effectively.
- Prototyping is one of the most critical steps in the design process, yet prototypes still confuse some designers and project teams.
- Prototypes differ in terms of their fidelity to the final product.
  - Low fidelity means prototype doesn’t include many detail
  - High fidelity prototype can be fully functional and behave



UI Design Process: Handing over to Developers

- When UX designers are fully satisfied with their designs, they create a document that has all the details and digital assets required for the development team to bring the product to life. This is the design handoff process.
- When a handoff is done correctly, it saves time and effort for both the developer and the designer, as well as improving the quality of the end product and the happiness of the user.
- The handoff documents bridge the gap between design and development and while it may sound like an overwhelming process, it doesn't have to be.
- To create a seamless design handoff for a developer, we share some important tips that UX designers should follow.

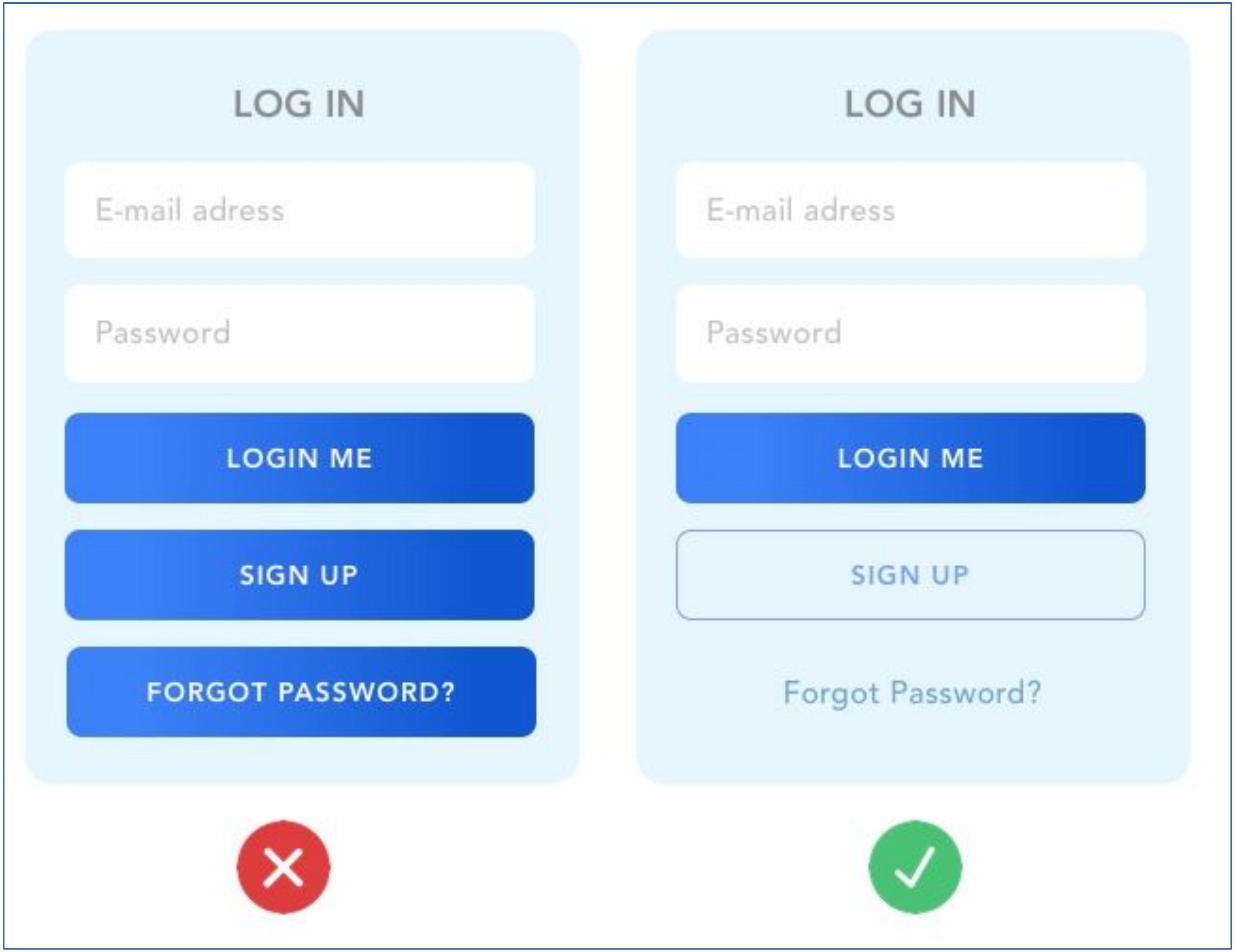


UI Design Process: Gathering User Feedback

- User feedback is essential for monitoring the health of your product and keeping track of your end users' needs.
- User feedback is, quite simply, feedback shared by people who use your product.
- It comprises suggestions, reviews, thoughts, opinions, and complaints—any insights your users or customers choose to share about their experience.
- User feedback is incredibly valuable because it comes from the people who matter most: your end users.
- It provides insight into how the product can be fixed or improved, what additional features could be added, and how satisfactory the user experience is overall.
- Feedback helps you to meet the needs and expectations of your target audience.
- This is essential for designing a successful, competitive product.

UI Design Issues

- **Inconsistent Design**: Inconsistent design elements, such as color palettes, typography, and iconography
- **Confusing Navigation**: Navigation that is difficult to use
- **Poor Layout**: A layout that is not well-organized
- **Slow Response Times**: Response times that are too slow
- **Ignoring User Needs**: Designing a product based on assumptions instead of user needs
- **Too Many Words**: Using too many words in the interface
- **Putting Style Over Substance**: Prioritizing style over usability
- **Following Design Trends Blindly:** Following design trends without considering the needs of the target users
- **Insufficient Feedback**: Not getting enough feedback from users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UX Design Principles

- **User-Centricity**

  - User-centricity, is to create products and services that solve user problems.
  - Everything a UX designer does should be steeped in the principle of user-centricity.
  - User-centricity means putting the user's needs first and making decisions based on what you know about them and what they want from the product.

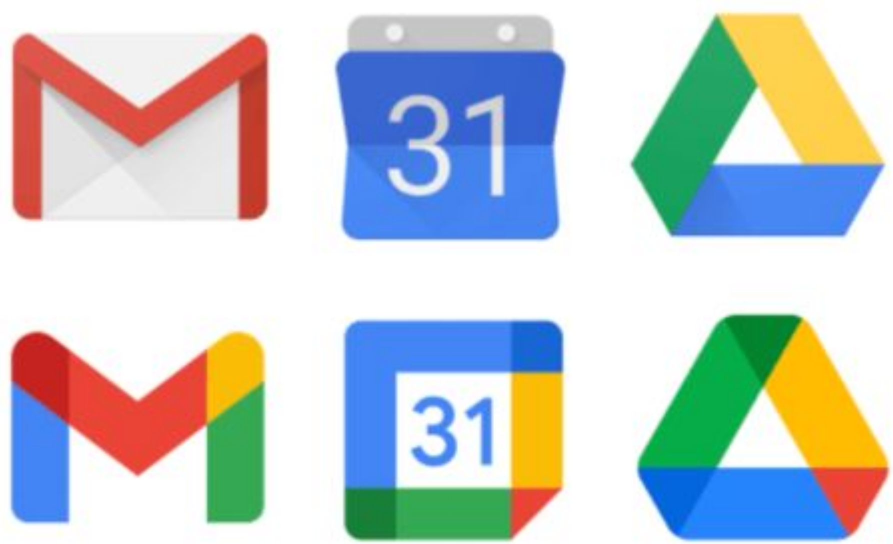
UX Design Principles

- **Consistency**

  - Consistency, is creating products that solve specific user problems, it’s essential to adhere to the UX design principle of consistency.

  - Whether it’s an app, a website, or a social media feed, there should be consistency in your visuals, such as buttons and links, and in your tone so, for example, your products are consistently business-like, irreverent, or playful.

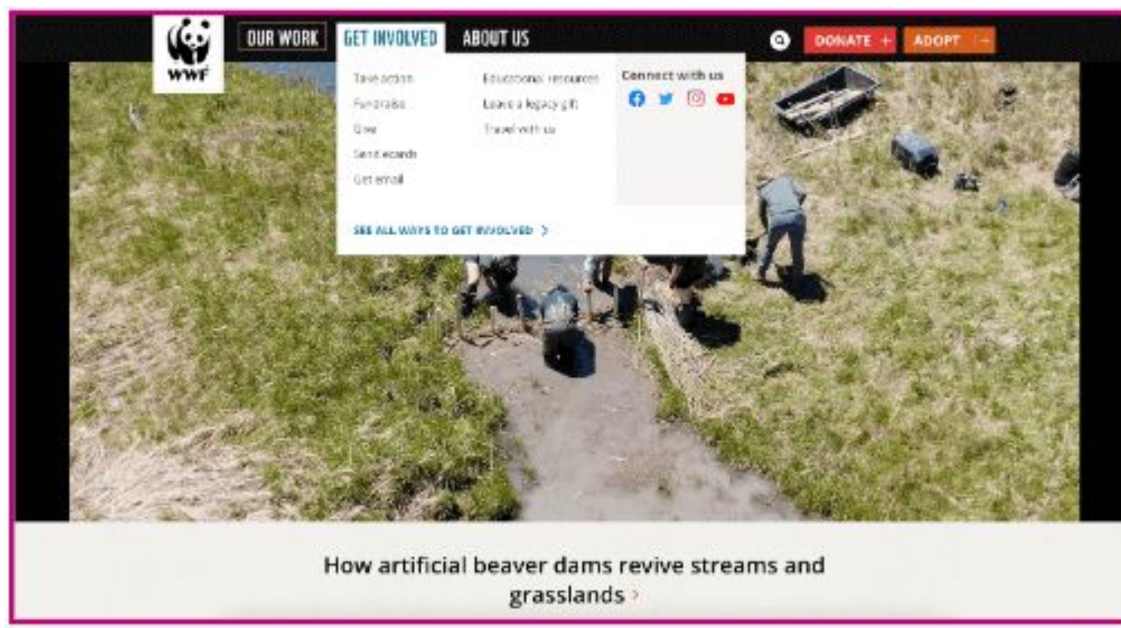
  - Consistency also extends to meeting the user’s expectations for the kind of product you’re designing.



UX Design Principles

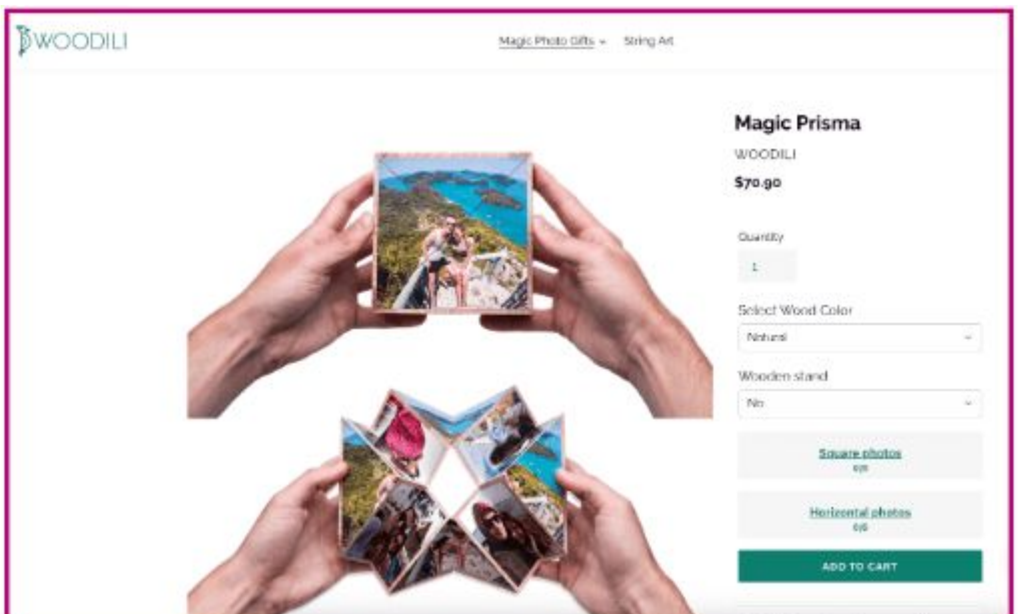
- **Hierarchy**

  - Hierarchy, is an important UX principle as it shapes how the user navigates a product — and how easy or complicated the process is.
  - Hierarchy relates to information architecture as well as the visual hierarchy of individual pages and screens.
  - More important elements can be emphasized by placing them at the top of the page or screen, using larger font or using different colours to help them stand out.
  - Hierarchy helps the user to navigate your product as it draws attention to the most important pages and elements, ensuring they can easily find what they need.

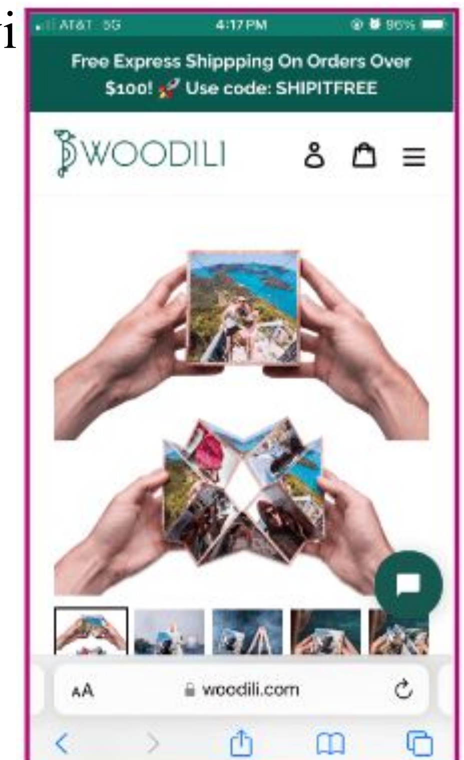


- **Context**

  - Context considers the circumstances in which your product will be used and how certain factors might impact the user experience.
  - Designer can ask questions as:
    - What device(s) might people use to access and interact with my product?
    - Where might the user be?
    - Are there factors such as noise which might interfere wi



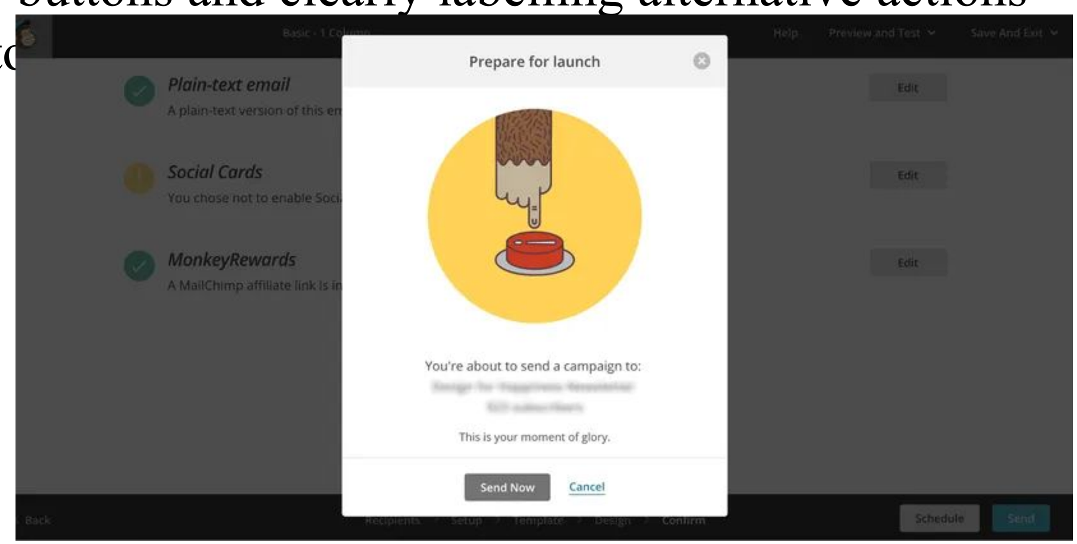
to be in?



UX Design Principles

- **User Control**

  - Users often perform actions by mistake.
  - They need a clearly marked ‘emergency exit’ to leave the unwanted action without having to go through an extended process.
  - There must be a balance between AI and user control, that gives users the ability to override AI decisions.
  - The principle of user control is all about helping users to correct or reverse errors without throwing the entire user experience into disarray.
  - You can build user control and freedom into your product by incorporating “Undo” and “Redo” functionality, providing “Cancel” buttons and clearly labelling alternative actions and routes the user can take if they want to

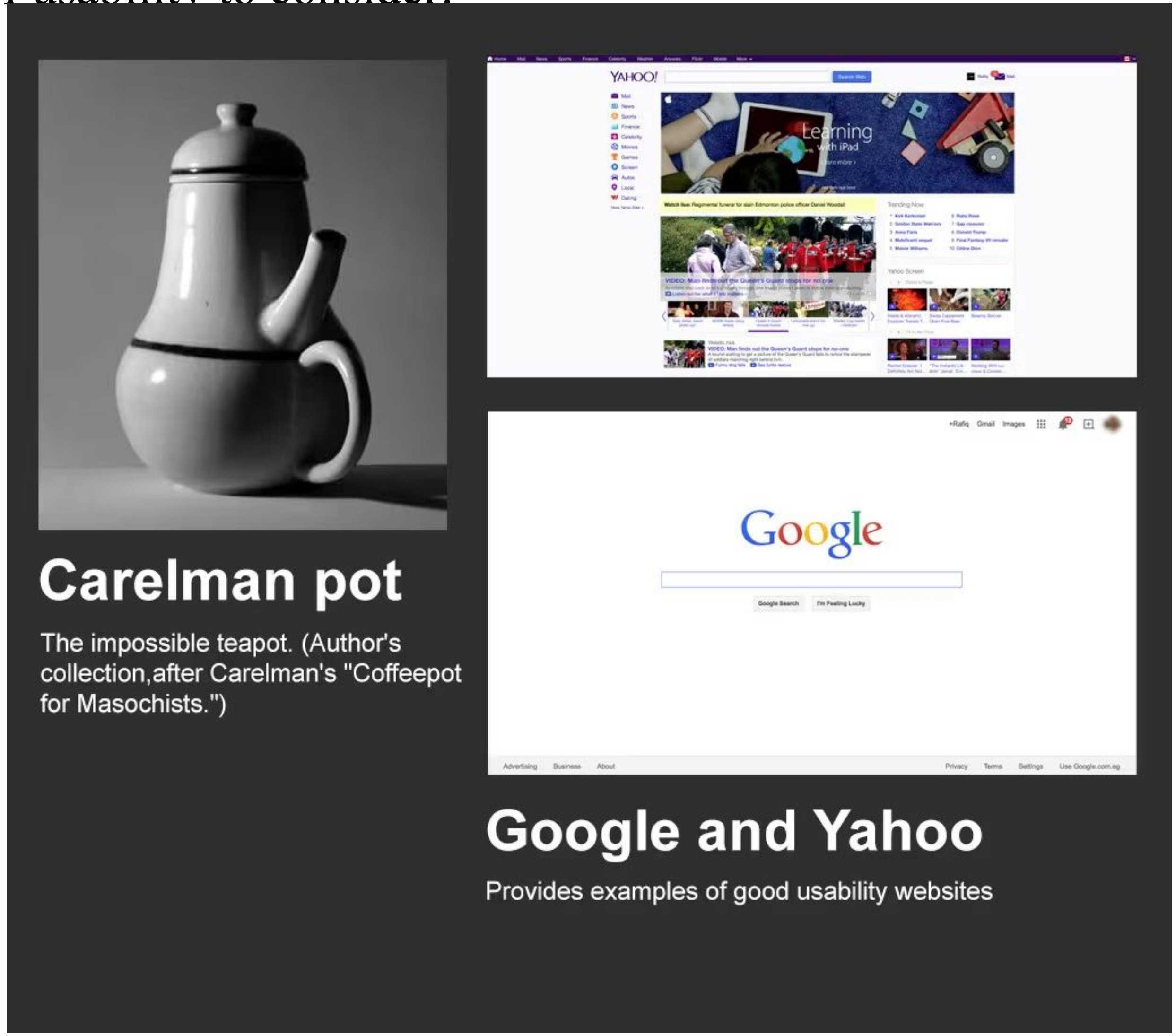


UX Design Principles

- **Accessibility**
  - Accessibility is about ensuring your product or service is accessible to and usable for as many people as possible.
  - It includes catering to the needs of people with disabilities, as well as understanding how different environments or situational factors might impact the user experience.
  - Example: Use high colour contrast to ensure that text is legible for users with visual impairments.

UX Design Principles

- **Usability**
  - Usability, is quite simply, a measure of how easy a product is to use.
  - There are five components of usability to consider:
    - Learnability
    - Efficiency
    - Memorability
    - Errors
    - Satisfaction



UX Design Process

- **Stage 1 – Project Definition and Scope**
  - The first stage of the UX design process is project definition and scope.
  - The purpose of this document is to provide a comprehensive overview of the project.
  - The project definition and scope will be defined, followed by a discussion of the project team and their qualifications.
  - Finally, the expected results of the project will be outlined.

UX Design Process

- **Stage 2 – Understanding the Problem**
  - The next stage of the UX design process is understanding the problem statement.
  - In order to improve a website, it is important to understand the problem.
  - This first step, known as problem discovery, allows designers to identify issues that users are having with the current design.
  - Once the problem is understood, the designer can begin to formulate a solution.
  - This process is known as problem solving and is an important part of the design process.

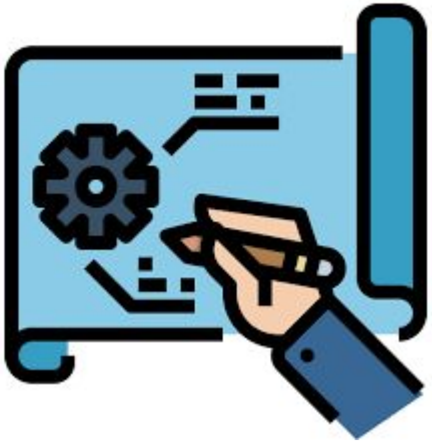
UX Design Process

- **Stage 3 - UX Research**
  - User experience (UX) research is an important part of any product development process.
  - It helps determine how well a product meets the needs of its users.
  - By understanding the needs of your target audience, you can create a product that is both user-friendly and attractive.
  - UX research can be divided into four main categories:
    - Demographic Research
    - Psychographic Research
    - Customer Interviews
    - Usability Testing

UX Design Process

- **Stage 4 – Ideation – Sketching and Low-Fidelity Prototyping**
  - The next stage of the UX design process is ideation.
  - When creating a new product, it is important not to get too caught up in the details.
  - The first step is to simply come up with ideas, which can be done through sketching and low-fidelity prototyping.
  - This allows for a broad range of ideas to be generated without getting bogged down by the details.
  - Once a general direction has been decided upon, then more detailed work can be done.

- **Stage 5 – High-Fidelity Mockups and Prototypes**
  - High-fidelity mockups and prototypes are an important part of the design process.
  - By definition, a mockup is a scale or full-size model of a design or device, used for demonstration, promotion, design evaluation, or other purpose.
  - In general, the main purpose of a mockup is to communicate how something will work.



UX Design Process

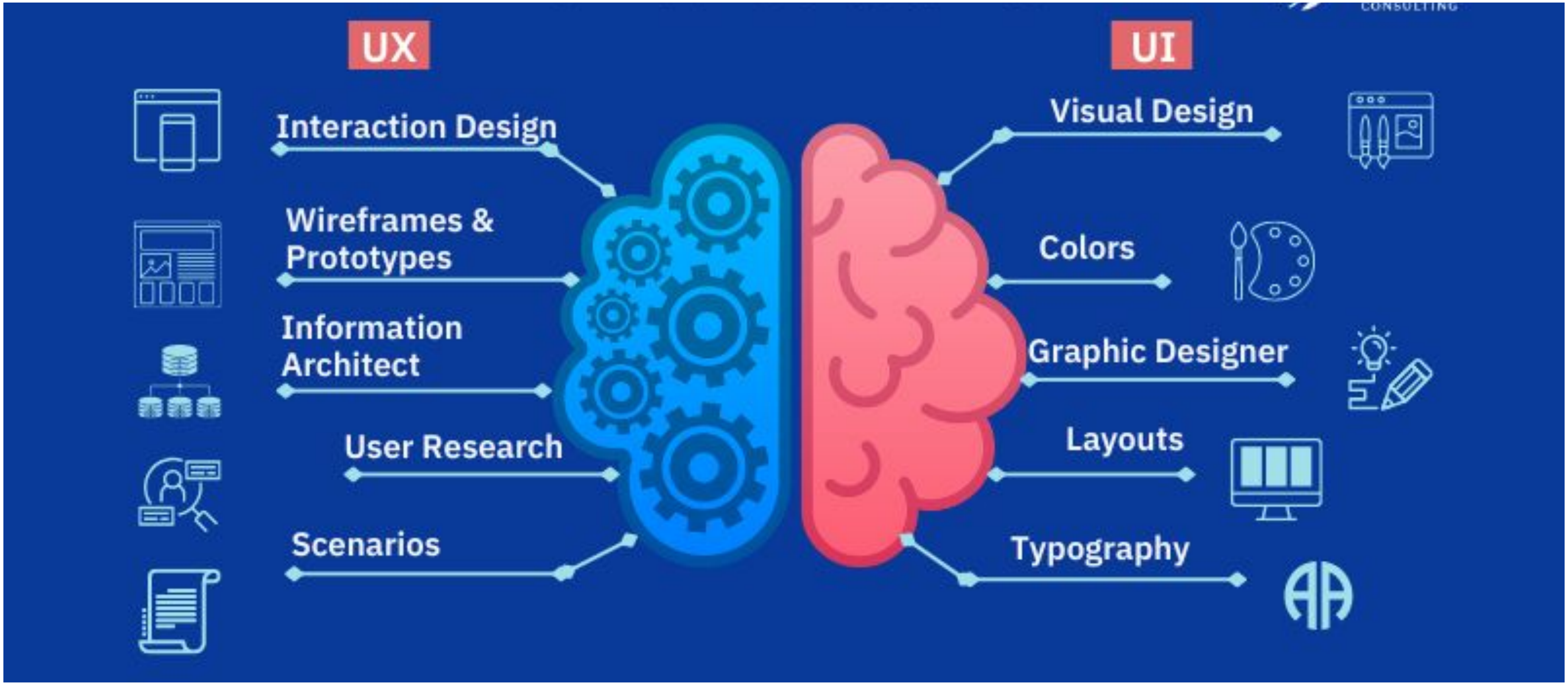
- **Stage 6 – Usability Testing**
- The next stage of the UX design process is usability testing.
- A prototype can be helpful in testing the usability of a design.
- By having an early version of the design that is interactive, you can test how people will interact with it.
- You can also use a prototype to test the design’s layout and appearance.

- **Stage 7 – Design Handoff**
- In order to successfully complete a design handoff, it is important to have a clear plan and a good communication strategy between the different team members.
- By having a plan in place, the team can avoid any misunderstandings or delays in the project timeline.
- Design handoff in UX design is important for improving user experience.
- By involving the design team early in the process, we can ensure that the user interface is well-considered from the get go.

UX Design Process

- **Stage 8 – Quality Assurance or UX Audit**
- The final stage of the UX design process is QA and UX audit.
- Quality Assurance (QA) generally refers to a planned and systematic process for ensuring that products (services and information) meet specified requirements and are fit for their intended purpose.
- In software development, it is often referred to as software testing.
- QA is used in all aspects of product development, from idea generation to customer feedback after product release

UI vs UX



UX
UI
Interaction Design
Wireframes & Prototypes
Information Architect
User Research
Scenarios
Visual Design
Colors
Graphic Designer
Layouts
Typography

Softare Coding Standards

- It refers to as well defined and common style of coding followed by several programmers working on same projects.
- It prescribes rules and guidelines about declaration and naming of variables, formatting features, etc
- Types of Coding Standards:
  - General coding standards
  - Language coding standards
  - Project specific coding standards
- **Advantages of Coding Standard:**
  - Gives uniform appearance to codes
  - Maintains Consistency throughout program
  - Makes code easier to understand for debugging and modification
  - Encourages good programming practices

Software Documentation Guidelines

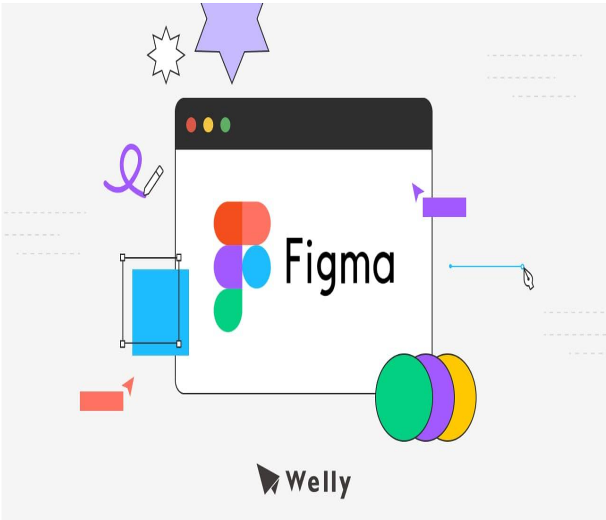
- Software documentation is a written piece of text that is often accompanied by a software program.
- It may contain anything from API documentation, build notes or just help content.
- It is a very critical process in software development.
- For a programmer reliable documentation is always a must the presence keeps track of all aspects of an application and helps in keeping the software updated.
- Types of Software Documentation:
  - **Requirement Documentation**: It is the description of how the software shall perform and which environment setup would be appropriate to have the best out of it. These are generated while the software is under development and is supplied to the tester groups too.
  - **Architectural Documentation:** Architecture documentation is a special type of documentation that concerns the design. It contains very little code and is more focused on the components of the system, their roles, and working. It also shows the data flow throughout the system.
  - **Technical Documentation:** These contain the technical aspects of the software like API, algorithms, etc. It is prepared mostly for software devs.
  - **End-user Documentation:** As the name suggests these are made for the end user. It contains support resources for the end user.

Introduction to FIGMA

Figma is a cloud-based **UI/UX design and prototyping tool** widely used in software engineering to design, collaborate on, and test user interfaces for web and mobile applications. It helps bridge the gap between **design and development** by allowing designers and engineers to work together in real time.

Figma enables teams to:

- Design **user interfaces (UI)**
- Create **interactive prototypes**
- Collaborate in real time through a web browser
- Share designs easily with developers and stakeholders



Unlike traditional design tools, Figma runs in the browser, which means **no installation is required** and designs are accessible from anywhere.

Why Figma is Important in Software Engineering

In software engineering, Figma plays a key role in the **design phase** of the Software Development Life Cycle (SDLC):

- Translates **requirements** into visual designs
- Improves communication between **developers, designers, and clients**
- Reduces rework by validating designs before coding
- Speeds up development with reusable components and design systems

Key Features of Figma

- UI Design Tools: Frames, shapes, text, colors, and layouts
- Prototyping: Create clickable, interactive flows without writing code
- Collaboration: Multiple users can edit and comment in real time
- Components & Design Systems: Reusable UI elements for consistency
- Developer Handoff: Inspect panel provides CSS, iOS, and Android specs
- Version History: Track and restore previous design versions

Figma in Agile Development

In **Agile and Scrum** environments, Figma supports:

- Rapid prototyping during sprint planning
- Continuous feedback from stakeholders
- Iterative design improvements aligned with development sprints

Advantages of Using Figma

- Platform-independent (works on Windows, macOS, Linux)
- Real-time collaboration
- Easy sharing and feedback
- Free plan suitable for students and small teams
- Integrates with tools like Jira, Slack, and GitHub